

HDFS Storage Layer

/crime_data/

18 CSV datasets + GeoJSON → Cleaned intermediate files → JSON analytics output

Stage 1: Data Preparation

src/data_preparation.py

Schema normalization • State name mapping (50+ variants) • Type casting • HDFS I/O

Stage 2: Analytics Engine

src/analytics.py

KMeans clustering • Ridge regression forecasting • Women safety index • Crime composition

Stage 3: Visualization

dashboard/generate_dashboard.py

Folium choropleth maps • Interactive trend charts • Standalone HTML dashboard

Shared Utilities

src/utils.py + src/state_mapping.py

Spark session management • HDFS helpers • State name canonicalization